

Question:

a) What is SecPM and why is it used? (2 Marks)

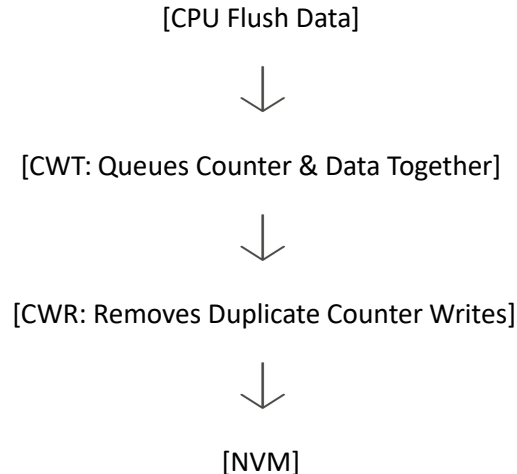
b) Explain the working principle of the CWT and CWR schemes in SecPM. (3 Marks)

Ans:

a) What is SecPM and why is it used?

SecPM is a hardware architecture for Non-Volatile Memory (NVM). It is used to provide **data security** (via counter-mode encryption) and ensure **crash consistency** without requiring software or compiler modifications.

b) Working of CWT and CWR Schemes:



1. Counter Cache Write-Through (CWT):

- When a cache line is flushed, the memory controller increments the encryption counter and writes it directly into the write queue alongside the encrypted data.
- Ensures data and counters are durable together, preventing data loss during a system crash.

2. Locality-Aware Counter Write Reduction (CWR):

- Exploits spatial locality where multiple counters share the same 64B memory line.
- Checks the write queue and removes redundant older counter entries with the same address.
- Cuts extra writes and increases speed.

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) **SecPM**

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SecPM NVM

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) **CWT** **CWR**

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[CPU Flush Data]



[CWT: Queues Counter & Data Together]



[CWR: Removes Duplicate Counter Writes]



[NVM]

. Counter Cache Write-Through (CWT):

○

(Write Queue)

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. Counter Write Reduction (CWR):

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